Toward a four-dimensional Yang–Mills construction

Notebook 20: the first Wilson small-field integration

Research working document

17 October 2026

Abstract

The physical source-coordinate construction of Notebook 19 and the overlap expansion of Notebook 18 yield an actual first-step small-field integration of the non-Abelian Wilson action. For the bare step, every action term is local before and after the exact pivot substitution. This verifies the support-valued precision and interaction hypotheses directly. The retained sharp cutoff is fixed; its nonlinear Jacobian, Haar density, gauge remainder and normalization terms are kept explicitly. The normalized fluctuation logarithm has connected activities whose anchored norm is bounded by $C[gR^3 + \exp(-cR^2)]$, for a logarithmic cutoff R. Constants are uniform in periodic volume at fixed blocking ratio. Strict support here is in the independent current background. Its transport to the next coarse field, the subsequent quantum iterations, and the complementary large-field regions remain the next construction steps.

This notebook applies the internally checked finite-regulator results in Notebooks 15–19. It is a proof of a specified small-field contribution. Completing the full existence argument requires that contribution to be incorporated into an iteration that controls the whole regulator measure and the continuum observable hierarchy.

1 An actual first-step Yang–Mills small-field integral

We now set the preceding depth to $M = 1$. The current classical action S_1 is the original, coupling-free Wilson action. This specialization supplies the interaction and precision-locality hypotheses directly: no localization theorem for a previously minimized action is needed. The block side L is fixed throughout. The conclusion is a convergent representation of the specified first-step small-field integral, with constants uniform in periodic volume. It does not include the complement of that integral or an iteration of quantum effective interactions.

Use the exact ordinary constraint, pivot chart Φ_V, and absolute contour gauge function $\mathcal{G}$ of Notebook 19. Set $a = \log V/i$ and write

$$\mathcal{F}_V(x) = S_1\big(\log(e^{i\Phi_V(x)}V)/i\big) + \mathcal{G}(e^{i\Phi_V(x)}V). \tag{1}$$

This symbol is the scalar action plus gauge function; it is not the group-valued coarse constraint. On the physical family $V = V_*(W)$ constructed in Notebook 19,

$$\mathcal{F}_V(0) = S_1(a), \qquad D\mathcal{F}_V(0) = 0, \qquad D^2\mathcal{F}_V(0) = A(V). \tag{2}$$

On a full product domain of independent small complex current link potentials, define $A(V) = D^2\mathcal{F}_V(0)$ using the actual absolute gauge function. This extension need not be stationary. Its flat value is $A_0 = K_L^T(q_1 + T_L^T T_L)K_L$ and has the spectral and exponential inverse bounds of Notebooks 17 and 19.

1.1 Finite local action and precision labels

Assign retained positive bonds to their L-blocks. There are $d(L^d-1)$ retained bonds per block in the stated periodic pivot geometry, so the number of real fluctuation coordinates per cell is the fixed integer $D = d(L^d-1)\dim \mathfrak{g}$. The background coordinates assigned to a cell include its pivot bonds as well. These background and fluctuation coordinate counts need not be equal. All supports below refer to these cells.

Each Wilson plaquette reads four current links. Substitution of $e^{i\Phi_V(x)}V$ enlarges this to a fixed neighborhood because each pivot correction reads one fixed local stencil. Each gauge contour likewise reads a fixed neighborhood. Hence there is an exact sum

$$\mathcal{F}_V(x) = \sum_r f_r(V|_{Y_r}, x|_{Y_r}), \tag{3}$$

with at most s_L cells in each Y_r, tree length at most ℓ_L, and at most n_L labels containing any given cell. Periodic occurrences are counted with their multiplicities. All f_r are holomorphic and uniformly bounded on a fixed larger background/fluctuation polydisc. The constants depend on L and the fixed gauge group, not on volume.

Local Cauchy estimates therefore prove, on a smaller polydisc,

$$A(V) = A_0 + \sum_r B_r(V|_{Y_r}), \tag{4}$$

$$B_r(V) = D_x^2 f_r(V,0) - D_x^2 f_r(1,0),$$

$$\nu_t(b) := \sup_i \sum_{r: i\in Y_r} \sum_{j\in Y_r} e^{t\ell(Y_r)} \|(B_r)_{ij}\|_{\|a\|_\infty<b} \le C_{L,t} b. \tag{5}$$

The matrices B_r are symmetric and supported in rows and columns in Y_r; they read only the background there. This is a support-valued precision decomposition, not merely a bound on the summed kernel. It retains every constraint and ordinary-coordinate curvature term automatically because it differentiates the full local functions in (3).

Define the third-and-higher action interaction on the product domain by

$$v_r^{\mathrm{act}}(V; g, z) = -g^{-2}\left[f_r(V, gz) - f_r(V,0) - gD_x f_r(V,0)z - \tfrac{1}{2}g^2 D_x^2 f_r(V,0)[z,z]\right]. \tag{6}$$

Its singularity at $g=0$ is removable. For $R \ge 1$ and $|g|R$ below a fixed radius, local Taylor estimates give

$$\sup_i \sum_{r: i\in Y_r} e^{a_0|Y_r| + \kappa\ell(Y_r)} \|v_r^{\mathrm{act}}\|_{b,R} \le C_{L,a_0,\kappa}|g|R^3. \tag{7}$$

Here the norm includes the independent background product ball and the complex fluctuation polydisc of radius R in each retained bond coordinate. One obtains the bound by applying one-variable Taylor's formula to $t \mapsto f_r(V, tz)$; the number of variables per label is bounded at fixed L. There is no sum over arbitrarily many fine variables: this section is specifically the bare $M=1$ step.

Add the normalized ambient straightening-Jacobian and Haar logarithms from Notebook 19, grouping all primitive terms with equal support. Denote the resulting interaction by v_Y. The gauge remainder is already part of (6) and is not added twice. For any fixed a_0, κ,

$$\sup_i \sum_{Y\ni i} e^{a_0|Y| + \kappa\ell(Y)} \|v_Y\|_{b,R} \le C_{L,a_0,\kappa}(|g|R^3 + |g|R + |g|^2R^2) \le C'_{L,a_0,\kappa}|g|R^3, \tag{8}$$

where $|g|R \le 1$. The same bound, up to a fixed factor, holds for $\|e^{v_Y} - 1\|$ when the displayed quantity is small. Every term is strictly local in the independent current background, even though substituting $V_*(W)$ need not preserve finite support in W.

1.2 The retained source cutoff fits the cell theorem

Let

$$\chi_R(z) = \prod_{b \text{ retained}} \mathbf{1}_{\{|z(b)|<R\}}, \tag{9}$$

where $|\cdot|$ is the real color norm. In each cell let χ_i be its product of these bond indicators. Then $1-\chi_i \le \mathbf{1}_{\{\|z_i\|_2 \ge R\}}$. Consequently the cell tail estimates of Notebook 18 also apply to these rectangles: their proofs only use a fixed real indicator, bounded by one, and the stated bound on its bad event. The interaction supremum is taken on the actual retained-bond rectangle. Fixed factors converting bond and cell norms are included in the constants in (8). No complex indicator is introduced. The cutoff is an exact pullback because $\Phi_V(gz)(b) = gz(b)$ on every retained bond.

1.3 Closing all smallness conditions in the first step

Put $C_0 = A_0^{-1}$. Choose $0 < \kappa < m$ with $\kappa + 2m$ below its established inverse-decay exponent and set $K = \|C_0\|_{\kappa+2m}$. Its real spectral bounds are $\lambda = B_*^{-1}$ and $\Lambda_0 = \gamma_*^{-1}$. Fix $0 < \theta \le 1/4$, for example $\theta = 1/(4D)$. The constants in Notebook 18's tunable strictly local theorem are now fixed independently of volume and of the coupling. Equation (5) permits a positive choice of b satisfying all its precision smallness conditions. Indeed its ν tends to zero with b, and all other constants at this stage are finite and fixed.

Let $R(g) = c(\log g^{-2})^p$, where $c > 0$, $p > 1/2$, and $g > 0$ is small. Take $a_0 = 4 + \log \mathcal{K}_\theta$ in (8), with the density constant of that theorem. Its interaction and bad-cell parameters obey

$$\epsilon(g) \le C_L g R(g)^3, \qquad \tau(g) \le C_L e^{-c_L R(g)^2}. \tag{10}$$

Both tend to zero. The recovered whole field stays inside the chart since $\|\Phi_V(gz)\|_\infty \le C_L g R(g)$. Thus a further, volume-independent coupling ceiling enforces the chart condition and both quarter budgets of Notebook 18. The choice of b precedes that ceiling; small coupling has not been used to repair a failed precision estimate.

Theorem 1.1 (The bare retained small-field integral)**.** *There are positive b, g_0, a_*, C, c_*, independent of compatible periodic volume, such that for $0 < g < g_0$ the function*

$$Z_g(V) = \int d\mu_{A(V)^{-1}}(z)\, \chi_{R(g)}(z) \exp\left(\sum_Y v_Y(V,z)\right) \tag{11}$$

has a holomorphic logarithm on the independent product ball $\|\log V/i\|_\infty < b$, and a convergent expansion

$$\log Z_g(V) = \sum_H E_{g,H}(V|_H), \qquad \sup_i \sum_{H \ni i} e^{a_*|H|} \|E_{g,H}\|_b \le C\{gR(g)^3 + e^{-c_* R(g)^2}\}. \tag{12}$$

The supports H are connected periodic cell sets. The same canonical expansion differs from that for the pure Gaussian cutoff by at most $CgR(g)^3$ in this norm.

Proof. The covariance baseline, support-valued precision, product domain, bounded local interaction multipliers and retained cutoff have all been checked above. Apply the tunable strictly local theorem and its periodic extension from Notebook 18. The right side follows from (10). The analytic continuation is that of the fixed real Gaussian integral, with its determinant branch chosen at the real positive baseline. It is not a distributional complex constraint integral. □

The product-domain extension subtracts the local linear terms in (6). On the physical stationary family their sum is zero by (2), so (11) is exactly the normalized fluctuation factor of the original action plus gauge. At nonstationary backgrounds it is the explicitly defined centered extension, not the unmodified Wilson integral.

1.4 The normalization terms also have local expansions

The delta-elimination factor is

$$\mathcal{D}(V)=\prod_c \det(A_{V,p(c)})^{-1}, \qquad \log\frac{\mathcal{D}(V)}{\mathcal{D}(1)}=-\sum_c \operatorname{Tr}\log\pi_c(V). \tag{13}$$

It is a sum of fixed-range local $O(b)$ terms. It has no coupling suppression and is not absorbed into (12). The Gaussian normalization similarly contributes

$$\mathcal{Q}(V)=-\tfrac{1}{2}\log\det(A(V)A_0^{-1})=\tfrac{1}{2}\sum_{n\ge 1}\frac{(-1)^n}{n}\operatorname{Tr}[(C_0B(V))^n], \qquad B(V)=\sum_r B_r(V). \tag{14}$$

The branch is zero at $V=1$.

For completeness, this determinant has a strict local expansion with volume-independent anchored $O(b)$ norm. Fix a small $t>0$ below the covariance exponent. A term in the expanded trace is a closed chain of covariance links and precision labels. Realize each covariance link by a chosen shortest periodic path and each label by a chosen minimal tree on Y_r. Their union H is connected and contains every background coordinate read by the term. If i is the initial trace cell and ℓ is the sum of these path/tree lengths, then

$$|H|\le 1+\ell, \qquad d(i,j)\le \ell \quad (j\in H). \tag{15}$$

Writing $K_t=\|C_0\|_t$ and using (5), the sum of absolute matrix products at order n, rooted at i and weighted by $e^{t\ell}$, is at most $D(K_t\nu_t(b))^n$. This follows successively from each weighted covariance row sum and then each weighted precision-label row sum; closing the chain only decreases the unrestricted endpoint sum. Choose $a_d,\beta>0$ with $a_d+\beta\le t$. For every $j\in H$, (15) gives $e^{a_d|H|+\beta d(i,j)}\le e^{a_d}e^{t\ell}$. Summing first over rooted chains and then their root yields

$$\sup_j\sum_{H\ni j} e^{a_d|H|}\|\mathcal{Q}_H\|_b\le\frac{De^{a_d}}{2}\left(\sum_i e^{-\beta d(i,j)}\right)[-\log(1-K_t\nu_t(b))]\le C_Lb. \tag{16}$$

The periodic lattice sum is bounded by its infinite-lattice counterpart. A further decrease of b ensures $K_t\nu_t(b)<1$. Absolute convergence justifies grouping by H and holomorphy of each local activity. The local delta terms in (13) meet the same type of bound.

1.5 Exact relation to the constrained group integral

This subsection makes the field-independent constants and the small-field restriction explicit. Let $q=\dim\mathfrak{g}$, let N_b be the number of current positive bonds, N_c the number of next-coarse positive bonds, and set $n=q(N_b-N_c)$. Choose Lebesgue coordinates orthonormal for the normalized trace and write normalized Haar measure near the identity as $d_H(e^{iX})=c_Hj(X)\,dX$, where $j(0)=1$. A normalized group delta at the identity consequently pulls back to $c_H^{-1}\delta(X)$.

For real physical $V=V_*(W)$ define $I_g^{\mathrm{sf}}(V)$ as the group integral of $\exp[-g^{-2}(S_1(U)+\mathcal{G}(U))]$ with constraint $\mathcal{M}_L(U)=W$, restricted to the local chart $U=e^{i\Phi_V(gz)}V$ and the retained rectangle (9). There are no other indicators in this definition. The exact ambient change of variables and pivot elimination from Notebook 19 give

$$\begin{aligned} I_g^{\mathrm{sf}}(V)=c_H^{N_b-N_c}g^n\mathcal{D}(1)(2\pi)^{n/2}(\det A_0)^{-1/2}\,e^{-S_1(a)/g^2}\\ \times\exp\left\{\log\frac{\mathcal{D}(V)}{\mathcal{D}(1)}+\mathcal{Q}(V)+\sum_H E_{g,H}(V|_H)\right\}. \end{aligned} \tag{17}$$

All terms on the second line have convergent strictly local current-background expansions. The classical term on the first line is kept explicit. At the physical stationary background it equals the next constrained minimum action by the minimum comparison in Notebook 19.

The exponential gauge can be inserted into a gauge-invariant constrained integral on a rooted-gauge-invariant admissible domain where the contour mean and ordinary average are defined and have their stated equivariance. This domain condition is needed for the source's logarithmic averaging maps; no global definition on every group configuration is assumed. The insertion has a known scalar normalization. Let

$$z_G(g) = \int_G d_H u \, \exp[-g^{-2}(1 - \operatorname{Re} \tau u)], \qquad N_s = \#\{\text{noncenter current sites}\}. \tag{18}$$

For rooted site gauges, contour equivariance gives $P_{U^u}(y,x) = P_U(y,x)u(x)^{-1}$. Haar invariance thus makes the gauge average of $e^{-g^{-2}\mathcal{G}(U^u)}$ exactly $z_G(g)^{N_s}$, independently of U in that domain. Averaging a gauge-invariant constrained integral over that domain proves the gauge-insertion identity with factor $z_G(g)^{-N_s}$. This identity holds before the chart restriction; the chart piece in (17) then has that same external factor. It does not equate a nongauge-invariant sharp chart piece before and after gauge insertion.

1.6 What this first step establishes

This is an actual non-Abelian Wilson small-field integration, including the source constraint geometry, fluctuation precision, retained cutoff, classical remainder, Haar density, straightening Jacobian, delta determinant and Gaussian determinant. Its bounds are uniform in periodic volume. It closes the first bare small-field step with no unknown coefficient hypotheses.

Three further passages are distinct. First, composing the strictly local current-background activities with $V_*(W)$ requires a locality theorem in the new coarse background. Second, subsequent steps must control the previously generated quantum interactions with uniform bounds. Notebook 21 supplies the intervening localization of the minimized classical action and one integration of it uniformly through earlier classical depth. Third, the regulator construction must control the complement of the chart, including compatible large-field histories, and produce the observable bounds needed for continuum reconstruction. These are not inferred from the first-step theorem.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 15–19 in this workspace. Notebook 18 proves the overlapping-interaction integration theorem and its strict support and periodic extensions. Notebook 19 gives the exact physical pivot chart, background Gaussian and local measure factors used here.

[2] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. **109** (1987), 249–301, original article copy.

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories. II. Cluster expansions*, Commun. Math. Phys. **116** (1988), 1–22, original article copy.

[4] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.